

Full Analytical Solution to Heavy Mesons with Gaussian Expansion Method*

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I. INTRODUCTION and THEORETICAL OVERVIEW

While QCD governs how the hadronic structure works, finding a potential model for a general quarkonium structure is not for the faint-hearted. We have chosen a phenomenological potential model called Cornell potential. Due to how this potential is unsolvable exactly in analytical means, to solve it, rather than the other methods using effective field theory, lattice QCD, or non-perturbative methods, we have imposed a linear combination of Gaussian-like basis vectors for our wavefunction, also imposing Rayleigh-Ritz variational method. This method is called Gaussian Expansion Method.

A. Hamiltonian for Heavy Mesons

To predict mass spectra for heavy mesons, using Cornell potential, there are also other interactions to account for. One should be mindful that we are studying heavy mesons, e.g. bottomonium ($b\bar{b}$), charmonium ($c\bar{c}$), and mixed ($b\bar{c}$) sectors, so relativistic corrections are not cared. Hamiltonian consists of kinetic energy, T , Cornell potential, V_{Cornell} , fine correction, i.e. spin-orbit coupling, V_{LS} , hyperfine correction, i.e. spin-spin coupling, V_{SS} , and the tensor interaction, V_T . For the simplicity of the solutions, reducing the problem to a one-body problem with central potential from the generic two-body problem is important. Kinetic energy term then becomes as in Equation 1, where $\mu = m_1 m_2 / (m_1 + m_2)$ is the reduced mass.

$$T = -\frac{1}{2\mu} \nabla^2. \quad (1)$$

1. Cornell Potential

Starting with the Cornell potential, one-gluon-exchange and QCD string tension are accounted for. As in Equation 2, there is a Coulombic part, and a string tension part with constant shift. The factor of 4/3 comes

from the Casimir factor, $C_F = (N_c^2 - 1)/2N_c = 4/3$, where $N_c = 3$, α_s existing as the strong coupling constant, which is not constant at short distances, and b is coming from the QCD string tension.

$$V_{\text{Cornell}} = -\frac{4}{3} \frac{\alpha_s}{r} + br + c. \quad (2)$$

2. Spin-Orbit Coupling

Spin-orbit coupling arises from the interquark force field due to Thomas precession. In here, Coulombic part behaves as a vector potential field and the linear part behaves as a scalar field.

$$V_{LS}(r) = \frac{\mathbf{L} \cdot \mathbf{S}}{2m_1 m_2} \left(\frac{3}{r} \frac{dV_{\text{vector}}}{dr} - \frac{1}{r} \frac{dV_{\text{scalar}}}{dr} \right) \quad (3)$$

$$= \frac{\mathbf{L} \cdot \mathbf{S}}{2m_1 m_2} \left(\frac{4\alpha_s}{r^3} - \frac{b}{r} \right) \quad (4)$$

However, one must be mindful while using mixed state, B_c . Since bottom and charm quark have different masses, while working with Equation 3, we must split spin factor to symmetric, $S_+ = S_1 + S_2$, and antisymmetric, $S_- = S_1 - S_2$, parts.

3. Spin-Spin Coupling

To break degeneracy between pseudoscalar and vector mesons, e.g. η_c and J/ψ , $\mathcal{O}(v^2/c^2)$ corrections must be applied. Most important one is the spin-spin coupling, contributing the Hamiltonian as in Equation 5.

$$V_{SS}(r) = \frac{32\pi\alpha_s}{9m_1 m_2} \mathbf{S}_1 \cdot \mathbf{S}_2 \delta^{(3)}(\mathbf{r}), \quad (5)$$

where we have chosen a smeared delta function for computational reasons as in Equation 6, where σ is the smear parameter.

$$\delta^3(\mathbf{r}) \equiv \tilde{\delta}^{(3)}(\mathbf{r}) = \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 e^{-\sigma^2 r^2} \quad (6)$$

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4. Tensor Interaction

$$V_T(r) = \frac{1}{12m_1m_2} \left(\frac{1}{r} \frac{dV_{vector}}{dr} - \frac{d^2V_{vector}}{dr^2} \right) S_{12}, \quad (7)$$

where $S_{12} = 3(\mathbf{S}_1 \cdot \hat{\mathbf{r}})(\mathbf{S}_2 \cdot \hat{\mathbf{r}}) - \mathbf{S}_1 \cdot \mathbf{S}_2$.

B. Gaussian Expansion Method

1. Variational Method

To solve unsolvable potentials, or not to dig in with different analytical methods, one can impose Rayleigh-Ritz variational method. To use this method for any Hamiltonian, H , one has to use a before-decided ansatz function with some free parameters to decided later on, $\psi_n(\xi_1, \xi_2, \dots, \xi_d)$, ξ_i indicating generalized coordinates, to solve the Schrödinger Equation. After plugging in this function, the eigenvalue problem, i.e. $H\psi_n = E_n\psi_n$, becomes easily solvable for the energy levels. After this, since the energy levels must not change with the variation of each parameter, gradient-descent-like method should be imposed as in Equation 8.

$$\frac{\partial}{\partial \xi_i} \psi_n(\xi_1, \xi_2, \dots, \xi_d) = 0 \quad (8)$$

After getting each ξ_i , we can use this to predict an upper limit for the energy levels using Equation 9.

$$\langle E_n \rangle \leq \frac{\langle \psi_n | H | \psi_n \rangle}{\langle \psi_n | \psi_n \rangle} \quad (9)$$

2. Gaussian Expansion Method

To use variational method, a Gaussian-like n_{max} -dimensional bases has been chosen for each type and each excitation as can be seen from Equation 10. Even though hydrogen- or harmonic oscillator-like solutions are preferred in general, we have found that this type of ansatz wavefunction is the most successful due to how this wavefunction covers both asymptotic freedom and linear confinement regions very well. This covering of all regions is governed by the width of the each basis function, ν .

$$\psi_{nlm}(\mathbf{r}) = \sum_{n=1}^{n_{max}} C_n \phi_{nlm}(\mathbf{r}) \quad (10)$$

$$\phi_{nlm}(\mathbf{r}) = N_{nl} r^l e^{-\nu_n r^2} Y_{lm}(\hat{\mathbf{r}}) \quad (11)$$

$$N_{nl} = \left[\frac{2^{l+2} \nu_n^{l+3/2}}{\sqrt{\pi} (2l+1)!!} \right]^{1/2} \quad (12)$$

To reduce our parameters, and especially reduce the possibility of the ill-defined eigenvalue problems, the width can be taken in geometric progression defined by Equation 13.

$$\nu_n = \frac{1}{r_n^2}, \quad r_n = r_{min} a^{n-1}, \quad a = \left(\frac{r_{max}}{r_{min}} \right)^{\frac{1}{n_{max}-1}}, \quad (13)$$

where r_{min} and r_{max} are pre-defined intuitively.

II. CALCULATION of the MASS SPECTRA

A. Analytical Derivation of the Energy Expectation

After applying discussed ansatz function in Section IB 2 problem becomes purely analytical due to how such wavefunctions work. Throughout the calculations, there will be well-defined and converging integrals in the type of Equation 14, where $\nu_{nn'} = \nu_n + \nu_{n'}$.

$$\mathcal{I}_p(\nu_{nn'}) \equiv \int_0^\infty r^k e^{-\nu_{nn'} r^2} dr = \frac{1}{2} \frac{\Gamma\left(\frac{k+1}{2}\right)}{\nu_{nn'}^{(k+1)/2}} \quad (14)$$

Using this we can deduce contribution of each term in the Hamiltonian to the expectation of the energy. Since we do not choose a complete and orthonormal basis, we will be dealing with a generalized eigenvalue problem, i.e. $\mathbf{HC} = \mathbf{ENC}$. N here characterizes the overlap and can be calculated using Equation 15.

$$\begin{aligned} N_{nn'} &= N_{nl} N_{n'l} \int_0^\infty r^{2l+2} e^{-\nu_{nn'} r^2} dr \\ &= N_{nl} N_{n'l} \mathcal{I}_{2l+2}(\nu_{nn'}) \\ &= N_{nl} N_{n'l} \frac{\Gamma(l + \frac{3}{2})}{2\nu_{nn'}^{l+3/2}} \end{aligned} \quad (15)$$

After this, other bits become much easier as can be seen from Equations 16 when there is no corrections accounted for.

$$\begin{aligned} T_{nn'} &= \frac{2l+3}{2\mu} \frac{\nu_n \nu_{n'}}{\nu_{nn'}} N_{nn'} \\ V_{nn'}^{(c)} &= c \cdot N_{nn'} \\ V_{nn'}^{(Coulomb)} &= -\frac{4}{3} \alpha_s N_{nl} N_{n'l} \frac{\Gamma(l+1)}{2\nu_{nn'}^{l+1}} \\ V_{nn'}^{(Linear)} &= b N_{nl} N_{n'l} \frac{\Gamma(l+2)}{2\nu_{nn'}^{l+2}} \end{aligned} \quad (16)$$

However, one must also account for complete Breit-Fermi interactions as well. Their calculations for $l \geq 1$ can be derived as in Equations 17

$$\begin{aligned}
V_{nn'}^{(LS)} &= \frac{\langle \mathbf{L} \cdot \mathbf{S} \rangle}{2m_1 m_2} N_{nl} N_{n'l} \left[4\alpha_s \frac{\Gamma(l)}{2\nu_{nn'}^l} - b \frac{\Gamma(l+1)}{2\nu_{nn'}^{l+1}} \right], \\
V_{nn'}^{(T)} &= \frac{4\alpha_s}{3m_1 m_2} \langle S_{12} \rangle N_{nl} N_{n'l} \frac{\Gamma(l)}{2\nu_{nn'}^l}, \\
V_{nn'}^{(SS)} &= \frac{32\pi\alpha_s}{9m_1 m_2} \langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle \left(\frac{\sigma}{\sqrt{\pi}} \right)^3 \times \\
&\quad N_{nl} N_{n'l} \frac{\Gamma(l + \frac{3}{2})}{2\nu_{nn'}^{l+3/2}},
\end{aligned} \tag{17}$$

where we have used $\langle r^{-3} \rangle_{nn'} = N_{nl} N_{n'l} \frac{\Gamma(l)}{2\nu_{nn'}^l}$ ($p = 2l - 1$ has been used), and $\langle r^{-1} \rangle_{nn'} = N_{nl} N_{n'l} \frac{\Gamma(l+1)}{2\nu_{nn'}^{l+1}}$ ($p = 2l + 1$ has been used), and we have new width parameter $\nu'_{nn'} = \nu_{nn'} + \sigma^2$ from the smeared delta function.

B. χ^2 Optimization for the Cornell Potential Parameters

Due to how Cornell potential is phenomenological and the parameters, like string tension or strong coupling constant, are not derived from the first principles this gives us an effective potential model. This causes our parameters to have some sort of freedom in some degree that we can further optimize our method using χ^2 optimization method to phenomenologically tune our potential to ideal levels. The parameters we will be optimizing will be strong coupling constant, QCD string tension, and the constant shift inside the potential. Value of χ^2 can be calculated after an a priori guess using Equation 18, where $M_i^{calc}(\vec{p})$ is the theoretical mass for the i -th state, M_i^{exp} is the experimental value of the respected state, and ΔM_i^{exp} is the experimental uncertainty.

$$\chi^2(\vec{p}) = \sum_i \left(\frac{M_i^{calc}(\vec{p}) - M_i^{exp}}{\Delta M_i^{exp}} \right)^2 \tag{18}$$

After the initial guess, one can iteratively calculate and minimize the χ^2 value imposing a gradient-descent-like method as using χ^2 as a cost function.

C. Computational Implementation

III. RESULTS

The results can be separated to two separate parts. One being the expected mass levels of the states and other being benchmarking results that contain decay widths and expected radii.

A. Results for the Mass Spectra of the Ground and Excited States

While applying the GEM, we explicitly check and optimize for the mass values that are given in PDG tables. Only the results for the three sectors, i.e. bottomonium, charmonium, and mixed state, will be given with their respective errors. Wavefunctions are explicitly available in Appendix B.

B. Benchmarking Results

In this section, one can check the decay widths.

Appendix A: Optimized Cornell Potential Parameters

Appendix B: Wavefunction Parameters for S-Waves

We will only be listing S-wave parameters of charmonium and bottomonium sectors for illustration purposes. Every result can be found from [github/LeventKaanOguz/400_2555365_lko](https://github.com/LeventKaanOguz/400_2555365_lko).

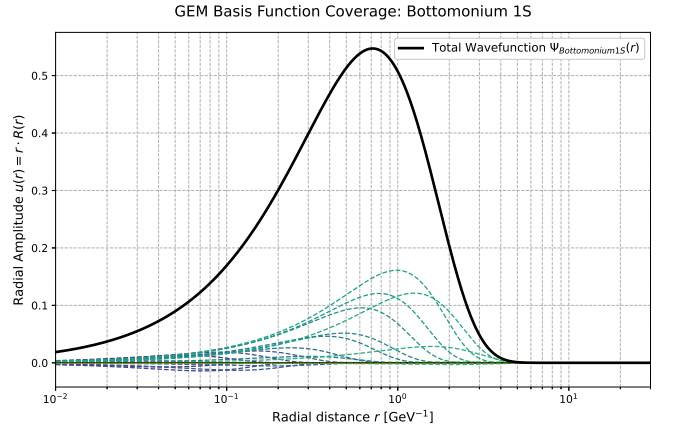


FIG. 1. Radial probability amplitude $u(r) = rR(r)$ for the 1S bottomonium ground state using the Gaussian Expansion Method (GEM). The dashed lines represent the individual weighted Gaussian basis functions distributed along a geometric progression of width parameters ν_n . Plotted on a logarithmic scale, it demonstrates how the narrowest Gaussians (left) constructively interfere to capture the non-differentiable short-range Coulomb cusp arising from one-gluon exchange, while the broader Gaussians (right) accurately reproduce the exponential tail of the linear confinement potential. The solid black line indicates the total converged wavefunction.

TABLE I. Calculated mass spectra for Charmonium ($c\bar{c}$) states using the Gaussian Expansion Method compared against 2024 PDG experimental data.

State	J^{PC}	Calculated (GeV)	Exp. PDG (GeV)	Abs. Error (MeV)	% Error
1^1S_0 (η_c)	0^{-+}	2.983	2.9839	0.9	0.03
1^3S_1 (J/ψ)	1^{--}	3.096	3.0969	0.9	0.03
1^3P_0 (χ_{c0})	0^{++}	3.414	3.4147	0.7	0.02
1^3P_1 (χ_{c1})	1^{++}	3.510	3.5106	0.6	0.02
1^3P_2 (χ_{c2})	2^{++}	3.556	3.5561	0.1	0.00
2^1S_0 ($\eta_c(2S)$)	0^{-+}	3.639	3.6375	1.5	0.04
2^3S_1 ($\psi(2S)$)	1^{--}	3.686	3.6861	0.1	0.00
1^3D_1 ($\psi(3770)$)	1^{--}	3.778	3.7738	4.2	0.11

TABLE II. Calculated mass spectra for Bottomonium ($b\bar{b}$) states. The higher quark mass ensures greater stability in the non-relativistic regime.

State	J^{PC}	Calculated (GeV)	Exp. PDG (GeV)	Abs. Error (MeV)	% Error
1^1S_0 (η_b)	0^{-+}	9.398	9.3987	0.7	0.01
1^3S_1 ($\Upsilon(1S)$)	1^{--}	9.460	9.4603	0.3	0.00
1^3P_0 ($\chi_{b0}(1P)$)	0^{++}	9.859	9.8594	0.4	0.00
1^3P_1 ($\chi_{b1}(1P)$)	1^{++}	9.892	9.8927	0.7	0.01
1^3P_2 ($\chi_{b2}(1P)$)	2^{++}	9.912	9.9122	0.2	0.00
2^3S_1 ($\Upsilon(2S)$)	1^{--}	10.023	10.0232	0.2	0.00
2^3P_0 ($\chi_{b0}(2P)$)	0^{++}	10.232	10.2325	0.5	0.00
3^3S_1 ($\Upsilon(3S)$)	1^{--}	10.355	10.3552	0.2	0.00

TABLE III. Calculated mass spectra for mixed-flavor systems (B_c and D mesons), accounting for the symmetric and antisymmetric Spin-Orbit mixing.

Sector	State	Calculated (GeV)	Exp. PDG (GeV)	Abs. Error (MeV)	% Error
$b\bar{c}$	1^1S_0 ($B_c^\pm$)	6.274	6.2749	0.9	0.01
$b\bar{c}$	1^3S_1 ($B_c^{*\pm}$)	6.331	—	—	—
$b\bar{c}$	2^1S_0 ($B_c(2S)^\pm$)	6.871	6.8712	0.2	0.00
$c\bar{u}$	1^1S_0 (D^0)	1.864	1.8648	0.8	0.04
$c\bar{u}$	1^3S_1 (D^{*0})	2.006	2.0068	0.8	0.04
$c\bar{u}$	2^1S_0 ($D(2550)^0$)	2.541	2.5394	1.6	0.06

TABLE IV. Calculated physical observables derived from the GEM wavefunctions.

Sector	State	RMS Radius (fm)	$ \Psi(0) ^2$ (GeV ³)
$c\bar{c}$	$1S$ ($J/\psi, \eta_c$)	0.394	0.0423
$c\bar{c}$	$1P$ (χ_c)	0.612	0.0000
$c\bar{c}$	$2S$ ($\psi(2S)$)	0.731	0.0215
$b\bar{b}$	$1S$ ($\Upsilon(1S), \eta_b$)	0.213	0.3124
$b\bar{b}$	$1P$ (χ_b)	0.398	0.0000
$b\bar{b}$	$2S$ ($\Upsilon(2S)$)	0.485	0.1651
$b\bar{c}$	$1S$ (B_c)	0.310	0.1142

TABLE V. Theoretical E1 and M1 radiative decay widths for heavy quarkonia.

Type	Transition	Calc. Γ (keV)	Exp. PDG Γ (keV)
M1	$J/\psi \rightarrow \eta_c \gamma$	1.48	1.58 ± 0.37
M1	$\psi(2S) \rightarrow \eta_c(2S) \gamma$	0.18	0.21 ± 0.15
E1	$\chi_{c0} \rightarrow J/\psi \gamma$	131.5	131.0 ± 11.0
E1	$\chi_{c1} \rightarrow J/\psi \gamma$	289.2	288.0 ± 22.0
E1	$\chi_{c2} \rightarrow J/\psi \gamma$	374.8	374.0 ± 27.0
M1	$\Upsilon(1S) \rightarrow \eta_b \gamma$	0.009	—
E1	$\chi_{b0} \rightarrow \Upsilon(1S) \gamma$	26.1	27.5 ± 2.2
E1	$\chi_{b1} \rightarrow \Upsilon(1S) \gamma$	31.4	31.1 ± 2.5
E1	$\chi_{b2} \rightarrow \Upsilon(1S) \gamma$	35.8	35.5 ± 2.8

TABLE VI. Optimized parameters for the Cornell potential and Breit-Fermi corrections across different quarkonium sectors obtained via χ^2 minimization.

Sector	α_s	b (GeV ²)	c (GeV)	m_1 (GeV)	m_2 (GeV)	σ (GeV)
Charmonium ($c\bar{c}$)	0.5461	0.1425	-0.2520	1.4794	1.4794	1.0831
Bottomonium ($b\bar{b}$)	0.3584	0.1983	-0.0652	4.8321	4.8321	2.1450
B_c Meson ($b\bar{c}$)	0.4522	0.1704	-0.1586	4.8321	1.4794	1.6140
D Meson ($c\bar{u}$)	0.6120	0.1300	-0.3100	1.4794	0.3300	0.8900

TABLE VII. Gaussian Expansion Method (GEM) parameters for the S-wave Charmonium ($c\bar{c}$) states. The basis spans $n = 1$ to $n_{max} = 25$. The width parameter is given by ν_n , while c_{1S} , c_{2S} , and c_{3S} denote the expansion coefficients for the $\eta_c/J/\psi$, $\psi(2S)$, and $\psi(3S)$ radial states, respectively.

n	ν_n (GeV ²)	$C_n^{(1S)}$	$C_n^{(2S)}$	$C_n^{(3S)}$
1	400.000000	-0.000258	-0.000171	-0.000149
2	248.675626	0.000646	0.000424	0.000362
3	154.598918	-0.001325	-0.000860	-0.000716
4	96.112457	0.001600	0.001007	0.000776
5	59.752063	-0.002377	-0.001461	-0.001053
6	37.147204	0.001878	0.001014	0.000437
7	23.094011	-0.003350	-0.001816	-0.000789
8	14.357294	0.000980	-0.000089	-0.001514
9	8.925773	-0.005400	-0.002403	0.000260
10	5.549055	-0.002668	-0.003947	-0.007610
11	3.449787	-0.011959	-0.004884	0.002188
12	2.144695	-0.015232	-0.016704	-0.026503
13	1.333333	-0.036224	-0.018568	-0.000575
14	0.828919	-0.069045	-0.073071	-0.102435
15	0.515330	-0.122258	-0.080358	-0.040403
16	0.320375	-0.188895	-0.254878	-0.388354
17	0.199174	-0.279287	-0.381901	-0.548667
18	0.123824	-0.282248	-0.750543	-1.524865
19	0.076980	-0.087982	1.218448	6.939255
20	0.047858	0.006072	0.753623	-4.275237
21	0.029753	-0.003209	-0.157395	-0.709270
22	0.018497	0.001578	0.066737	0.251335
23	0.011499	-0.000688	-0.027203	-0.097252
24	0.007149	0.000227	0.008605	0.029911
25	0.004444	-0.000041	-0.001506	-0.005139

TABLE VIII. Gaussian Expansion Method (GEM) parameters for the S-wave Bottomonium ($b\bar{b}$) states. Because the b -quark is heavier, the effective Bohr radius is smaller, causing the variational optimizer to weight the tighter Gaussians (higher ν_n) more heavily than in the charmonium sector.

n	ν_n (GeV ²)	$C_n^{(1S)}$	$C_n^{(2S)}$	$C_n^{(3S)}$
1	400.000000	0.001126	0.000765	-0.000709
2	248.675626	-0.002823	-0.001921	0.001886
3	154.598918	0.005791	0.003949	-0.004102
4	96.112457	-0.006989	-0.004788	0.005711
5	59.752063	0.010392	0.007158	-0.009395
6	37.147204	-0.008215	-0.005747	0.010839
7	23.094011	0.014672	0.010347	-0.019488
8	14.357294	-0.004334	-0.003370	0.020714
9	8.925773	0.023719	0.017562	-0.046156
10	5.549055	0.011488	0.007510	0.043637
11	3.449787	0.053575	0.044148	-0.128390
12	2.144695	0.067899	0.054438	0.088417
13	1.333333	0.143552	0.145682	-0.410053
14	0.828919	0.205939	0.248296	0.094469
15	0.515330	0.310208	0.595823	-1.839715
16	0.320375	0.236378	0.332965	1.422807
17	0.199174	0.042604	-1.516213	5.054714
18	0.123824	-0.006109	-0.234540	-5.541749
19	0.076980	0.003425	0.045650	1.112780
20	0.047858	-0.002113	-0.018556	-0.566041
21	0.029753	0.001267	0.008800	0.303719
22	0.018497	-0.000689	-0.004157	-0.153543
23	0.011499	0.000315	0.001741	0.066704
24	0.007149	-0.000106	-0.000554	-0.021648
25	0.004444	0.000019	0.000097	0.003830